Fusurge: An Accelerated Query-Driven System for Multimodal Information Retrieval

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Abstract. With the rapid expansion of multimedia archives, video retrieval systems must strike a balance between scalability, accuracy, and responsiveness. `Fusurge`, reported by TKU@MBZUAI in AI Challange HCMC 2025, solved this challenge by integrating a compact yet powerful pipeline. Key components include data processing for rapid keyframe sampling, PaddleOCR for multilingual OCR, and faster-whisper for efficient ASR. On the semantic side, multiple CLIP-based encoders are fused to widen coverage, while compact vision–language models support interactive question answering. Retrieval quality is further refined through reranking with user-guided clarification. An intuitive interface lowers entry barriers and ensures ease of use. Empirically, `Fusurge` demonstrates strong retrieval quality and robust system performance, thereby achieving consistent top-rank effectiveness up to 86% with responsive, stable runtime—making it suitable for real-world, large-scale use.

Keywords: Retrieval System · Query-based Retrieval · Interactive Retrieval.

1 Introduction

As video becomes a dominant medium for digital interaction, the demand for high-performance video retrieval systems is rapidly accelerating. Within the Ho Chi Minh AI Challenge [4] framework, we focus on realistic, interactive scenarios across three core tasks: Known-Item Search (KIS), Video Question Answering (QA), and TRAKE (Tracking Across Events). KIS is further categorized into two modes: t-KIS, which leverages natural language queries, and v-KIS, which relies on visual descriptors. Video QA introduces added complexity, requiring systems to generate accurate, contextually grounded responses directly from video content. Beyond these, our platform enables tracking-based retrieval for TRAKE tasks, where queries involve following specific entities or actions across multiple

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scenes and temporal segments. Our dataset comprises about 220GB of data, segmented into two batches of videos and keyframes, including per-video IDs and per-frame annotations.

To address the core challenge of the competition, `Fusurge` handles user queries over large-scale video repositories and returns the $top - k$ most relevant results. The system operates through a three-stage pipeline comprising data preparation, a retrieval backend, and a frontend visualization. Videos are segmented into keyframes, with multi-modal features extracted from visual, audio, and temporal domains. These features are indexed using Elasticsearch for structured and textual data and our rebuilt Milvus framework for vector-based similarity search. The back-end, built with FastAPI, handles query processing and retrieval, while the ReactJS and TailwindCSS front-end render results at reduced resolution for faster interaction. Ultimately, this robust infrastructure facilitates a user-centric design that supports complex queries combining textual descriptions, visual cues, and object attributes, empowering users to refine results with the high precision required for real-world video retrieval.

2 Related Work

Vision–Language Models Vision–Language Models (VLMs) encode visual and linguistic inputs into a joint multimodal embedding space [10]. Prominent architectures include OpenAI's CLIP, which projects image–text pairs into a shared space for efficient cross-modal retrieval [17], and InternVL, a state-of-the-art model that aligns visual inputs with linguistic reasoning to handle complex multimodal tasks [1].

Temporal Grounding Temporal grounding aligns natural language queries with specific video segments. This capability is central to the TRAKE tasks in the Ho Chi Minh AI Challenge [4], enabling the precise retrieval of sequential events. Consequently, it has become a key tool for top teams in competitions like the Video Browser Showdown (Fustar [11], Fusionista [12], [9]).

Search Methods: Two common search backends used in video retrieval systems are Milvus and Elasticsearch. Milvus is a high-performance vector database that embeds user queries into high-dimensional vectors and efficiently computes similarity against pre-stored text–image embeddings to identify the most relevant video segments. In contrast, Elasticsearch supports robust full-text queries, operates on non-vector data, and relies on keyword-based indexing, offering fast retrieval for textual data.

3 Data Preparation and Feature Extraction

To enhance the performance and reliability of downstream tasks in the Ho Chi Minh AI Challenge [4] competition, we construct a dedicated data preparation

pipeline to extract high-quality keyframes from raw video inputs. This pipeline comprises three main stages:

Visual Feature encoding: Each video is first encoded using the CLIP ViT B-32 [17] model to extract frame-level visual embeddings. This step transforms raw pixel data into a dense feature representation aligned with multimodal semantics, which serves as a foundation for downstream selection and understanding tasks.

Scene Boundary Detection: In parallel, we use TransNetV2 [21], a compact model for scene boundary detection, to segment the video into structurally coherent scenes. This imposes high-level temporal structure by grouping visually and semantically continuous frames, thereby improving the contextual precision of subsequent keyframe clustering.

Keyframe Selection: Using CLIP-based visual embeddings and detected scene boundaries, we apply an enhanced K-means clustering scheme that merges only temporally adjacent clusters with the closest centroids to select representative keyframes for each scene. The resulting keyframes form a compact yet informative visual summary and are stored for downstream use.

4 System Overview

4.1 Architecture

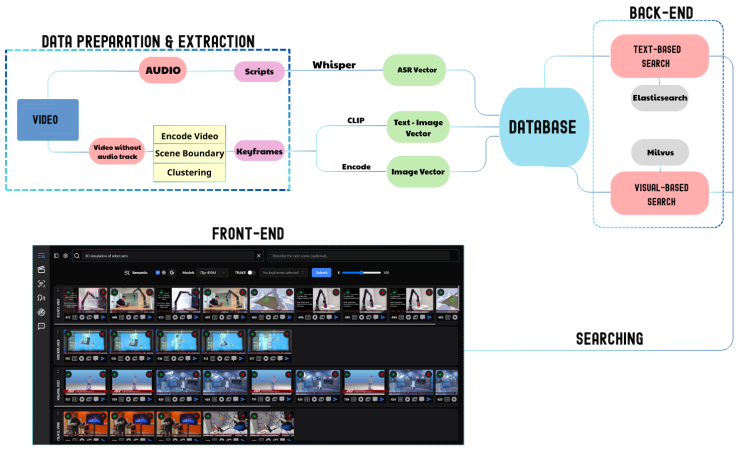


Fig. 1. Overview of Fusurge System

4.2 Functional Modules

Semantic Search via Query Decomposition. `Fusurge` leverages two compact yet state-of-the-art models: ViT-H-14-378-quickgelu and ViT-SO400M-14-SigLIP-384 [17]. Both models encode images and queries into a shared semantic space for cosine-similarity comparison, and the resulting embeddings are indexed in our rebuilt Milvus engine, enabling `Fusurge` to retrieve diverse and highly relevant results. Despite being under 1 billion parameters, these models achieve top-tier performance across 38 benchmark datasets [7]. Although CLIP performs well for image retrieval, it struggles with long queries. To address this, we break a user's input into multiple smaller queries using GPT-4o [14] and compute similarity with an exponential mean:

$$\text{Score}(Q, I) = \frac{1}{|Q|} \sum_{q \in Q} \exp\big(\text{sim}(q, I)\big) \text{ [12]}$$

where Q is the set of queries and I is the image. Here, $\text{sim}(q, I)$ denotes the similarity between query q and image I. The exponential mean behaves like a softmax, emphasizing images that strongly match at least one query while still rewarding those that match several queries.

Temporal Query Mechanism. Conventional semantic search methods treat frames as independent images and disregard their temporal dependencies, which severely limits event-level video retrieval where ordering and timing are crucial. In contrast, `Fusurge` consumes a sequence of temporally ordered queries, each tied to a specific event moment, and retrieves the video whose frame sequence best matches this query sequence, together with the aligned frames.

$$Score(Q, V) = \max_{(i_0, i_1, \ldots, i_k) \in \mathcal{I}} \left(\sum_{j=0}^{k} sim(Q_j, V_{i_j}) \right) \text{ [12]}$$

Here, $Q = (Q_0, Q_1, \ldots, Q_k)$ denotes the sequence of queries, $V = (V_1, V_2, \ldots, V_n)$ the sequence of video frames, $\mathcal{I}$ the set of all index tuples with $1 \leq i_0 < i_1 < \ldots < i_k \leq n$, and $sim(Q_j, V_{i_j})$ the similarity between the j-th query and the i_j-th frame. `Fusurge` selects the frame combination with the maximum total similarity, computed efficiently via dynamic programming in $\mathcal{O}(n \cdot f)$ time, where n is the number of frames and f the number of queries [12]. Through query decomposition, complex natural-language inputs are mapped to temporally grounded subqueries, enabling retrieval of *top-k* videos that are both semantically relevant and temporally consistent with the described event.

OCR Search. To construct a tool for recognizing text in images and videos, we developed a platform that integrates PaddleOCR [3] with VietOCR [16]. The processing workflow is organized into two primary stages: (i) detection and

precise refinement of text regions using the PP-OCRv5-server module of PaddleOCR [3], which is optimized for both accuracy and speed while maintaining lightweight deployment, as illustrated in Table 1, and (ii) recognition of textual content using VietOCR [16].

Table 1. PP-OCRv5 vs. Large-Scale Models: Accuracy and Size Comparison [3].

Model	1-EditDist	Params (B)
PP-OCRv5	**0.804**	**0.07**
Qwen2.5-VL-72B	0.801	72
Gemini-0605	0.781	80
InternVL2-78B	0.768	78
Kimir-VL-A3B	0.744	3
RoflmOCR-7B	0.732	7

VietOCR [16] is built upon the integration of CNNs with Language Models, specifically Seq2Seq [22] and Transformer architectures, trained on over 1 million synthetic image–text pairs to handle diverse fonts, layouts, and noise. The combination of these models shows high effectiveness on the competition dataset, achieving strong accuracy and robustness in real-world conditions, as illustrated in Figure 2.

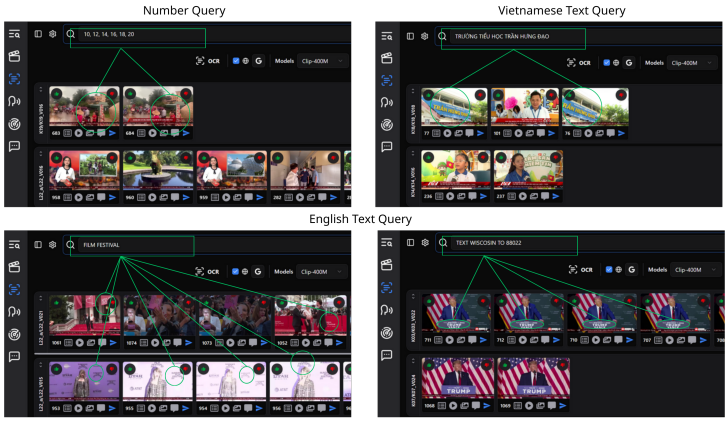


Fig. 2. Querying via OCR across multiple query types.

ASR Search. We uses faster-whisper [23] to convert audio into text. It is a lightweight and optimized version of OpenAI's Whisper, offering faster inference and high accuracy [23], making it suitable for real-time or large-scale processing. The audio is first transcribed into text using faster-whisper, with precise word-level timestamps. The transcript is then normalized, tokenized and aligned to

video keyframes based on timing to enable full-text search across audio and video.

Object Detection. YOLOv8 [24] is employed for object detection, as it provides a strong trade-off among accuracy, speed, and complexity. During data processing, we extract feature vectors for object identity, spatial location, and color, and organize them into a 10×10 grid, enabling queries over 80 predefined object categories combined with 16 colors. Inspired by Fusionista [12], we integrate Elasticsearch with K-means clustering to quantize pixel colors, obtain dominant colors, and encode them consistently with object features. For each user query, the canvas sketch is converted into a complete query representation and encoded for efficient retrieval in Elasticsearch, as illustrated in Figure 3.

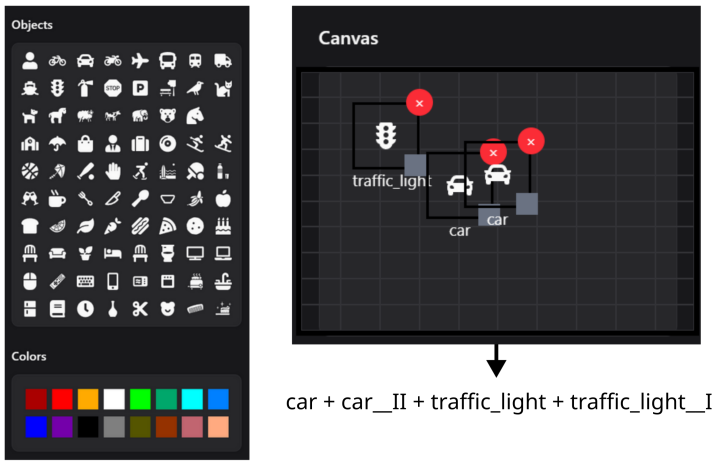


Fig. 3. Object detection based on shape, color, and position.

Question Answering. While multimodal large models such as GPT-4V [15] and MultiMood [10] achieve strong VQA performance [13], they are less suitable for time-critical, resource-constrained deployments due to API latency and external dependency (GPT-4V) or large parameter counts and heavy GPU requirements (LLaVA). We therefore adopt InternVL [1], a 1B–7B parameter family that attains competitive accuracy on benchmarks such as VQAv2 [6] and TextVQA [20], while offering faster and more predictable inference, making it a practical backbone for our QA module in the Ho Chi Minh AI Challenge [4].

Reranking Approach. To refine retrieval results, we employ a two-stage reranking pipeline, following the idea of [5], [8]. Given a potentially long or

complex input query, GPT-4o [14] first reformulates it into short concise yes/no verification questions that capture its key aspects [8] (e.g., "Is there a person", "Is the umbrella red?"). Candidate frames and these questions are then evaluated by a vision-language model, which outputs binary responses used to rescore and reorder the candidates. Among models in the 1B-7B range, we adopt InternVL3.5-4B [1] as the reranker, as it achieves the best trade-off between efficiency and accuracy under strict latency constraints.

4.3 UI/UX Design

`Fusurge` provides a web-based application built with React and TailwindCSS on top of a FastAPI backend, designed to balance accessibility with the efficiency required in competitive retrieval scenarios. The interface is structured around six panels that correspond to the supported modalities—semantic search, temporal search, OCR, ASR, object-based retrieval, and question answering. Each panel is equipped with modality-specific controls that enable users to refine queries according to their needs, including a query-decomposition mechanism powered by GPT-4o [14] for handling long or complex inputs.

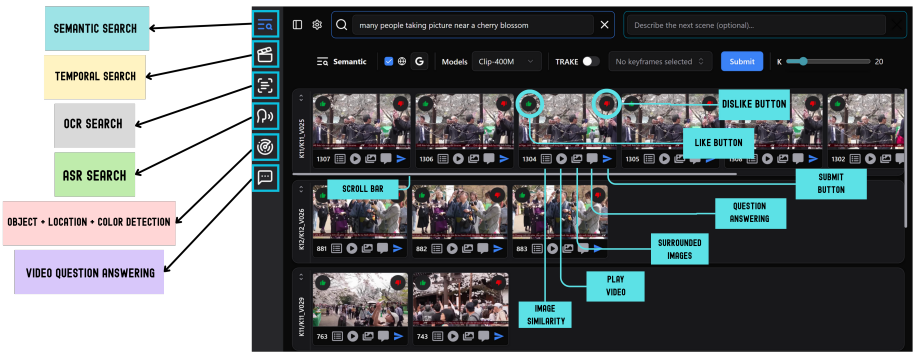


Fig. 4. Main System Interface.

Our interface builds on the Fustar [11] design and extends it with advanced video-querying tools and intelligent shortcut mechanisms that enhance efficiency and user experience (Figure 4). To ensure that retrieved results remain relevant, a reranking component based on InternVL3.5-4B [1] is integrated directly into the interface. User interactions such as "like" and "dislike" automatically adjust the ranking of results. In addition, the system supports a variety of convenient interaction features: retrieved frames are accompanied by contextual utilities, including direct video playback, navigation to neighboring frames, similarity-based result expansion, transcript alignment for ASR outputs, and a scroll bar beneath the results area that enables efficient browsing of large result sets.

5 Experiments

5.1 Dataset

Collected in two stages, the dataset encompasses approximately 330 hours of audiovisual material distributed across 1478 video files. Instead of focusing on a single type of content, it includes a mixture of program formats such as broadcasts, performances, instructional material, and domain-specific shows, ensuring broad coverage.

5.2 Evaluation Metrics

The Ho Chi Minh AI Challenge 2025 [4] is evaluated in two distinct stages: the Qualification Stage and the Grand Final.

Qualification Stage. The first stage evaluates ranked retrieval with up to 100 answers per query, measuring performance by the Mean Top-k R-Score $k \in \{1, 5, 20, 50, 100\}$ to capture top accuracy and depth performance..

The R-Score serves as the core evaluation metric, capturing how well a prediction aligns with the ground truth. For Visual Question Answering (VQA), the prediction must additionally provide the correct textual answer:

$$R\text{-Score}(r_i) = \mathbb{I}(v_i = GT_v \ \wedge \ id_i \in [s, e] \ \wedge \ a_i = GT_a)\,. \tag{1}$$

For Temporal Alignment (TRAKE), correctness is more gradual: if the video ID is correct, the score is the fraction of predicted segments that fall inside the ground-truth interval, and otherwise the score is zero:

$$R\text{-Score}(r_i) = \begin{cases} \dfrac{1}{N}\displaystyle\sum_{j=1}^{N} \mathbb{I}(id_{i,j} \in [s, e])\,, & \text{if } v_i = GT_v, \\ 0, & \text{if } v_i \neq GT_v. \end{cases} \tag{2}$$

The final query score is the average of these values across the five cutoffs, computed by dividing their sum by 5.

Grand Final Round. The Grand Final shifts focus from ranked retrieval to interactive search under time pressure, adopting the standard evaluation protocol of the VBS. Each query can yield a maximum of 100 points, subject to a time limit t_L. The scoring function for a query is:

$$f_{\text{KIS}}(t, w_s) \ = \ \left\lceil \max\left\{0, \ 50 + 50 \times f_{\text{TS}}(t) \ - \ 10 \times w_s\right\}\right\rceil, \tag{3}$$

where t is the response time ($0 \leq t \leq t_L$) and w_s is the number of incorrect submissions. The design of this formula reflects two competing pressures: rapid answers are rewarded through the time-dependent term $f_{\text{TS}}(t)$, while careless guessing is discouraged by penalizing wrong attempts. The overall score is the sum across all queries, with real-time submission and scoring managed by DRES [19].

5.3 Experimental Design and Results

To make our study reproducible and visually grounded, we deployed a self-hosted, DRES - compatible system that manages data, enforces evaluation protocols, and performs scoring. We then evaluated the system on **four custom tests** closely mirroring past the Ho Chi Minh AI Challenge [4] formats. For each test, we exercised the full retrieval pipeline to answer queries end to end. Quantitative results are reported in Table 2, showing a high early-hit rate (*Correct@10* $= 1.0$) [18] and strong reciprocal-rank consistency (*MRR@10* ≈ 0.86) [2], together with low operational latency (SRT $= 0.3$–$2\,\mathrm{s}$; KDL $< 1\,\mathrm{s}$) and a low error rate ($\leq 0.5\%$). Overall, users can quickly find a useful candidate in the first screenful of results, lowering time-to-answer and interaction cost.

Table 2. Overview of results: *Retrieval Quality* (left) and *System Performance* (right)

K	**Correct@K**	**MRR@K**
1	0.80	0.80
5	0.90	0.844
10	**1.00**	**0.86**

Metric	**Value**
Search Response Time	0.5–2 s
Keyframe Display Latency	$< 1\,\mathrm{s}$
Error Rate	$\leq 0.5\%$

Given that the competition query formats closely resembled prior editions, we adopted existing query-type categorization strategies and focused on improving generalization and reducing latency across system components. This yielded competitive performance in the Preliminary Round of the Ho Chi Minh AI Challenge [4], where **TKU@MBZUAI** achieved scores of 22.8, 29.0, and 33.6, totaling 85.4 out of 88.

In the Final Round, the system demonstrated high interactive efficiency. For both t-KIS and v-KIS, scores consistently exceeded 90, with a maximum of 98 and minimum response times below 10 seconds. Video QA results remained stable at approximately 90 points, typically requiring only 1–2 queries to localize the target segment and produce the grounded answer.

Overall, CLIP-based models yielded the strongest performance, especially when provided with richer textual descriptions. A translation API and robust OCR ensured accessible retrieval for Vietnamese text across diverse fonts, while a lightweight ASR module enabled reliable voice-based video queries, particularly for news. Temporal Search further improved multi-event retrieval by capturing complete event sequences and validating the most relevant results.

6 Conclusion

We introduced `Fusurge`, a video retrieval system built for the Ho Chi Minh AI Challenge [4]. By integrating optimized preprocessing, multilingual OCR, efficient ASR, and fused CLIP-based encoders, `Fusurge` achieves faster retrieval with improved accuracy and usability. Results from the qualification round confirm its ability to reduce latency, improve precision, and support interactive search, highlighting its potential for large-scale, real-world deployments.

Acknowledgments. This research was supported by The VNUHCM-University of Information Technology's Scientific Research Support Fund, and the Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE. We would like to express our sincere gratitude to our collaborators at Taekwondo UIT Club for their valuable contributions.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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